Experiment No. 26

Prolog Program to Sum the Integers from 1 to N (AI)

Aim

To write a Prolog program to compute the sum of integers from 1 to a given number N using recursion.

Objectives

- To understand recursive predicate definitions in Prolog.
- To use base and recursive facts/rules to compute a running sum.
- To use arithmetic evaluation with the 'is' operator.
- To display the computed result using format/2.

Algorithm

1. Start.
2. Define the base case: sum(0, 0).
3. Define the recursive rule: sum(N, Sum) computes sum(N-1, Sum1) and adds N to Sum1.
4. Query sum with the desired value of N.
5. Prolog recursively reduces N until it reaches the base case.
6. The results are accumulated back up the recursive calls.
7. Display the final computed sum.
8. Stop.

Source Code (Prolog)

```
% Prolog program to sum integers from 1 to N
sum(0, 0).
sum(N, Sum) :-
    N > 0,
    N1 is N - 1,
    sum(N1, Sum1),
    Sum is Sum1 + N.
:- sum(10, Result),
   format("Sum of integers from 1 to 10 = ~w~n", [Result]).
```

Output

```
Sum of integers from 1 to 10 = 55
```

Result

The Prolog program successfully computed the sum of integers from 1 to 10 using recursion, giving the correct result of 55.